

D-Band RX Front-End With a 0° – 360° Phase Shifter Based on Programmable Passive Networks in SiGe-BiCMOS

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Abstract—Active phased arrays are key enablers for high-capacity wireless links and imaging sensors at millimeter wave, but require advanced front-end circuits. On the receiver side, the front-end comprises a low-noise amplifier (LNA) followed by a programmable phase shifter (PS), required to adjust the phase of each channel before performing the coherent summation of the signals captured by different antenna elements. In D-band, conventional PSs based on the vector interpolation principle limit the dynamic range with a noise or linearity penalty due to transistors operating close to $f_{\max}$, in currently available silicon technologies. This work presents a front-end where the variable phase shift is achieved using passive structures, with a noise figure equal to the insertion loss (IL) but inherently linear. Different passive networks providing a programmable phase shift in fine and coarse steps are developed and interleaved with active gain stages to build a 0° – 360° PS. Cascode structures are used as gain stages, in the PS and in the preceding LNA, and reactive feedback is introduced around the common-emitter (CE) device to boost the gain. A D-band receiver front-end is implemented in BiCMOS 55-nm technology. With a power consumption of 80 mW from a 2 V supply, measurements prove 20 dB average gain, 130–170 GHz operating frequency with 0° – 360° phase shift control, and average NF and $OP_{1\text{dB}}$ of 7 dB and -2 dBm, respectively. Normalizing the dynamic range to power consumption, the achieved results compare favorably against state-of-the-art.

Index Terms—BiCMOS, front-end, low-noise amplifier (LNA), phase shifter (PS), phased array, SiGe.

I. INTRODUCTION

THE never ending demand for faster transfer rate in mobile applications and ubiquitous connectivity is posing severe

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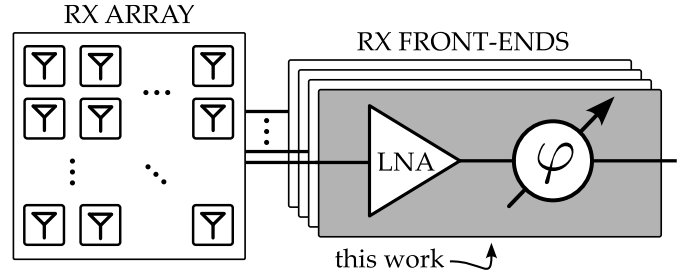


Fig. 1. Block diagram of a phased-array receiver front-end.

requirements to the mobile network infrastructure. To address this challenge, the wide bandwidth available beyond 100 GHz opens to wireless links with unprecedented capacity, and significant spectrum portions have already been reserved in the D-band (130–170 GHz) for point-to-point links in 5G and beyond [1]. Exploitation of the same band is also being actively investigated for the evolution of radar sensors, driven by the advantage on resolution provided by the short wavelength and the wide bandwidth [2].

For such kind of applications, BiCMOS technology gives an optimal compromise between cost and performance, joining high-speed and high-power HBTs with CMOS nodes, thus empowering the integration of systems on chip (SoC) in the sub-THz band. To cope with the high path loss, transceivers exploit phased-array architectures, where multiple radiating elements rise the equivalent antenna gain through beamforming. In active beamforming systems [3], [4], [5], [6], the radiating elements are grouped and connected to multiple power amplifiers (PAs) or low-noise amplifiers (LNAs) at the TX and RX side, respectively, and are followed by a programmable phase shifter (PS). In this way, the system performance is enhanced by a higher radiated power and better sensitivity. Moreover, active phased arrays benefit from the flexibility given by a programmable beam-steering capability [7].

This work focuses on a receiver front-end for a D-band active phased array, shown in Fig. 1. The front-end provides low-noise amplification and 0° – 360° programmable phase shift, required for the coherent combination of multiple channels, and its noise and large-signal compression are key not to compromise the system dynamic range. Several works have demonstrated the feasibility of such a component [8], [9] but,

in currently available technologies with transistors working close their frequency limit ($f_{\max}$), achieving low noise and high linearity at low power consumption remains challenging. The PS following the LNA is critical. The most common PS architecture is based on the vector interpolation principle, where a hybrid coupler generates quadrature signals which are then weighted and combined by a pair of variable-gain amplifiers to introduce the desired phase shift [10], [11], [12]. This architecture suffers from low gain (or loss) in the D-band, limited linearity, even at a moderately high power consumption, and poor noise figure (up to 20 dB, as in [13]). The PS noise may be suppressed with high gain from the preceding LNA, at the cost of linearity degradation.

Most of the D-band front-ends previously reported display an output-referred 1 dB gain compression point ($OP_{1\text{dB}}$) in the range from -10 to -5 dBm with a power consumption of 50–100 mW. Nevertheless, a higher compression point is often desirable, in radar sensors to withstand strong close-in reflections, but also in wireless communications. As an example, the system considerations in [1] suggest an input-referred 1 dB gain compression point ($IP_{1\text{dB}}$) of -20 dBm for a 256-QAM D-band link, corresponding to $OP_{1\text{dB}} = 0$ dBm with a typical 20-dB front-end gain. This compression level is demonstrated by the front-end in [8] with a power consumption of 165 mW, subsequently integrated into the multichannel active phased array presented in [14].

To mitigate the noise–linearity tradeoff, this work investigates a front-end where the variable phase shift is introduced by passive structures, inherently linear. Phased-array front-ends with passive PSs were successfully demonstrated at lower frequency, well below $f_{\max}$, although with limited phase resolution [15], [16]. To date, a limited number of works explored programmable passive PSs in sub-THz bands, particularly in silicon. A reflective-type PS with fine-control phase steps was investigated in [17], but demonstrated narrowband operation. On the other hand, [18] and [19] proposed switched transmission lines (Tlines) above 100 GHz in silicon and GaAs, respectively, but they require a large area that does not fit a sub-THz phased-array system. The compact passive networks used in this work, initially presented in [20], provide a programmable phase shift in fine and coarse steps. This extended study elaborates an optimal arrangement of those blocks, with a particular focus on enhancing the front-end performance, and finally demonstrates a high dynamic range D-band RX phased-array front-end. To this aim, the passive phase shifting networks are interleaved with active gain stages to build a 0° – 360° PS. Cascode structures are used as gain stages and gain-boosting is performed using reactive feedback around the common-emitter (CE). The effectiveness of the approach was previously experimented in a two-stage D-band LNA [21] and the technique is here adopted in the complete front-end. A test chip is realized in a SiGe BiCMOS technology by STMicroelectronics and draws 80 mW from a 2 V supply. The measured gain is 20 dB across the 130–170 GHz band and the phase shift is programmable over 360° with 7° resolution. The noise figure of 7 dB is aligned or better than previous works in the same band, while $OP_{1\text{dB}}$, from -3.5 to -2 dBm,

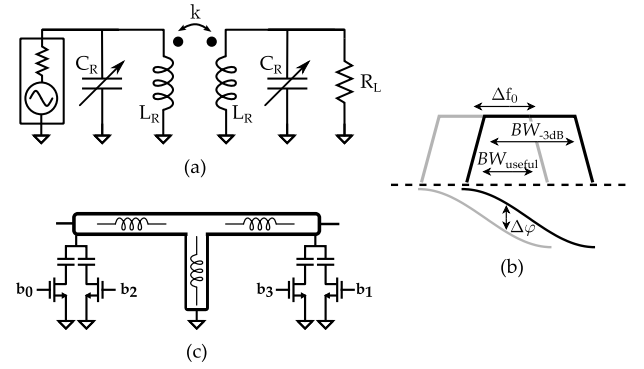


Fig. 2. (a) Coupled-resonator bandpass filter and (b) transfer function (amplitude/phase) variation by tuning the center frequency. (c) Schematic of the implemented network.

is remarkably higher than what previously demonstrated at comparable power consumption.

This article is organized as follows. Section II presents the different passive networks developed to provide variable phase shift in fine and coarse control steps. Section III analyzes the performance of a cascode stage in D-band and discusses the gain-boosting technique adopted in this work, while the design of the front-end is described in Section IV. The measurement results and a performance comparison with previous works are reported in Section V, followed by Conclusion.

II. PROGRAMMABLE PASSIVE PSS

Different passive networks providing a programmable phase shift are presented in this section. To cover the four quadrants (0° – 360°) with enough resolution, phase shifting networks with fine and coarse control are developed. The former feature a digitally controlled phase shift in small steps but the maximum range is limited to keep the insertion loss (IL) low. The latter provide a 1-bit phase control with a large step (nominally $0^\circ/180^\circ$ and $0^\circ/90^\circ$). By cascading the fine- and coarse-control networks, the four quadrants are covered maintaining fine resolution and limited IL.

A. Tunable Bandpass Filter for Fine-Control Phase Shift

The fine-control PS network is based on a fourth-order bandpass filter with tunable center frequency, similar to what is proposed in [22]. Such kind of frequency response may be achieved with magnetically coupled resonators, shown in Fig. 2(a). An exhaustive analysis of this network with several design guidelines is available in [23]. If the components (R , L , C , k) are sized such that the two pairs of complex conjugate poles provide a flat in-band response, the phase of the transfer function is fairly linear across frequency, with a variation of π radians within the -3 dB bandwidth, $BW_{-3\text{dB}} = f_H - f_L$ [Fig. 2(b)]. Therefore, $d\phi/df \approx \pi/BW_{-3\text{dB}}$. If the filter center frequency, f_0 , is shifted by Δf_0 by varying the network capacitance [gray plot in Fig. 2(b)], the signal experiences a phase shift variation equal to

$$\Delta\phi \approx \frac{\pi}{BW_{-3\text{dB}}} \cdot \Delta f_0. \quad (1)$$

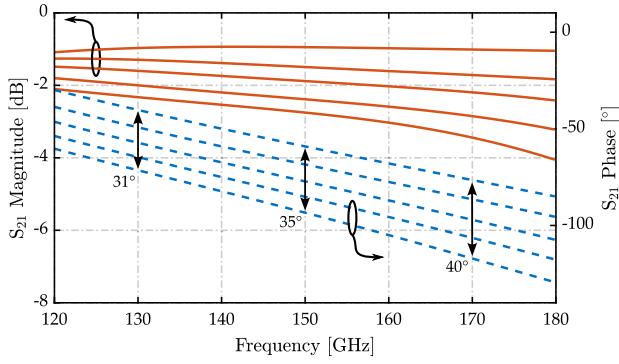


Fig. 3. S_{21} magnitude (solid) and phase (dashed) of the tunable filter at different states.

However, to avoid attenuation when the network transfer function is translated, the useful bandwidth (BW_{useful}) is lower than $BW_{-3\text{dB}}$. Looking at Fig. 2(b), $BW_{\text{useful}} = BW_{-3\text{dB}} - \Delta f_0$, and, making use of (1)

$$BW_{\text{useful}} = BW_{-3\text{dB}} \left(1 - \frac{\Delta\varphi}{\pi} \right). \quad (2)$$

From (2), given a target BW_{useful} , the wider the desired maximum phase variation, the larger must be $BW_{-3\text{dB}}$. On the other hand, $BW_{-3\text{dB}} \propto 1/(RC)$, and thus an upper bound on $BW_{-3\text{dB}}$ (and hence $\Delta\varphi$) is finally set by the smallest variable capacitor that can be reliably implemented to achieve the required tunability of f_0 . To extend the attainable phase shift, multiple identical networks can be cascaded; this leads to higher out-of-band roll-off with only a marginal bandwidth reduction. The implemented schematic is shown in Fig. 2(c). Differently than in the implementation of [22], where coil-based transformers are used, here the coupled inductors of Fig. 2(a) are replaced by their equivalent T-network, made with short Tlines, thus avoiding the high losses that affect magnetically coupled transformers at high frequency. The Tlines realize shielded microstrips with the signal routed in the topmost metal layer and display a quality factor $Q_{\text{TL}} = 35$ at 165 GHz. The network center frequency is digitally tuned by a bank of four switched MOM capacitors. The switches are nMOS with $W/L = 40 \mu\text{m}/55 \text{ nm}$, implemented in triple well to improve the substrate isolation and reduce losses [24]. Each MOM-switch combination features $C_{\text{MAX}} = 16 \text{ fF}$ with $C_{\text{MAX}}/C_{\text{min}} = 1.5$ and minimum quality factor $Q_{\text{min}} = 10.5$. The bandpass cells are sized for a center frequency of 150 GHz and 90% fractional bandwidth. The capacitors bank shifts the center frequency by $\pm 10\%$, thus, using (2), each cell introduces roughly 35° . Fig. 3 shows S_{21} magnitude and phase of the unitary cell. The total IL ranges between 0.9 and 3.2 dB, while the phase shift variation is 31° and 40° at 130 and 170 GHz, respectively, with an average phase step of 7° . The input and output return losses (not shown) are above 15 dB over the 130–170 GHz frequency range. The core size of the tunable filter is $90 \mu\text{m} \times 85 \mu\text{m}$.

B. 0°/180° and 0°/90° PSs

The reflective-type PS architecture is selected to implement a 0°/180° phase shift network. The schematic is reported

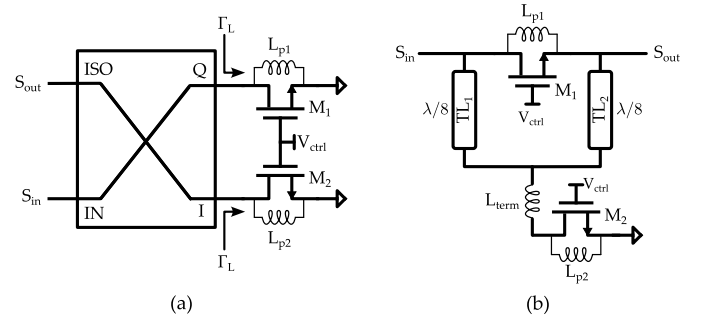


Fig. 4. Schematic of (a) 0°/180° hybrid-coupler-based PS and (b) 0°/90° switched-Tline-based PS.

in Fig. 4(a) and consists of a quadrature hybrid whose I and Q ports are terminated onto variable impedances. The input signal S_{in} splits evenly and appears halved at the I/Q ports. If an identical impedance Z_L with nonzero reflection coefficient (Γ_L) loads the I/Q ports, the signals are reflected and get summed at the ISO port, giving the output signal S_{out} . Neglecting the hybrid losses, the following relation holds:

$$\left| \frac{S_{\text{out}}}{S_{\text{in}}} \right| = |\Gamma_L|^2 \quad \angle \frac{S_{\text{out}}}{S_{\text{in}}} = -90^\circ + \angle \Gamma_L. \quad (3)$$

The I/Q ports are terminated by the nMOS switches $M_{1,2}$, yielding ideally $\Gamma_L = \pm 1$ in the OFF-state and ON-state, respectively. Using (3), S_{in} is transferred to the ISO port with the same magnitude but $\pm 90^\circ$ phase, corresponding to a 180° relative phase variation. The finite channel resistance and the parasitic capacitance of the nMOS switches introduce signal loss and deviation from the ideal 180° phase. The transistors are sized with $W/L = 50 \mu\text{m}/55 \text{ nm}$ to achieve $r_{\text{ON}} = 7 \Omega$. In the OFF-state, the parasitic capacitance at the drain terminal is resonated out by the parallel inductors $L_{p1,2} = 40 \text{ pH}$, implemented with meandered Tline stubs, to rise the equivalent OFF-state impedance to 250Ω . By doing so, $|\Gamma_{L,\text{ON}}| \approx |\Gamma_{L,\text{OFF}}|$, and the magnitude of S_{21} of the network is equalized between the ON-state and OFF-state. The simulated S_{21} with $V_{\text{ctrl}} = 0 \text{ V}$ and $V_{\text{ctrl}} = 1.2 \text{ V}$ is reported in Fig. 5. The relative phase change is 148° at 130 GHz and 200° at 170 GHz. The IL is between 3 and 4.4 dB. The input/output return loss (not shown) remains above 16 dB over the full bandwidth. The hybrid coupler determines the occupied silicon area. It is realized with $230 \mu\text{m}$ -long coupled lines with a total size of $250 \mu\text{m} \times 120 \mu\text{m}$.

The hybrid coupler could also be used to implement a 0°/90° PS by selecting with a switch either the I or the Q output, but with an intrinsic IL of 3 dB because half of the signal power is wasted [19]. To avoid this loss, the proposed 0°/90° PS leverages a bypassed $\lambda/4$ -long transmission line. The schematic is reported in Fig. 4(b). Inductors $L_{p1,2} = 60 \text{ pH}$, in parallel to the nMOS switches $M_{1,2}$, resonate the transistors parasitic capacitors to rise the OFF-state impedance and improve isolation. In this condition, L_{term} is floating and the signal flows through TL_1 and TL_2 , experiencing a 90° phase shift due to the $\lambda/4$ total length. When $M_{1,2}$ are ON, instead, the Tlines are bypassed by M_1 and the signal experiences ideally no phase shift. To limit the IL and improve the return

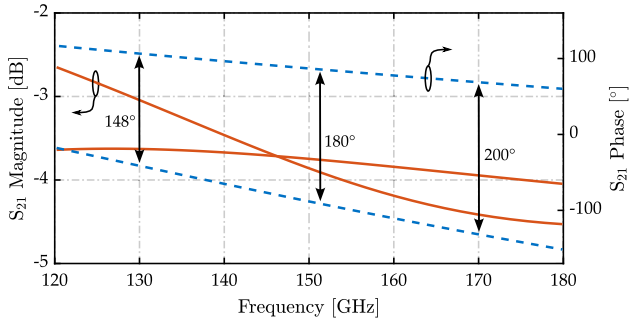


Fig. 5. S_{21} magnitude (solid) and phase (dashed) of the reflective-type $0^\circ/180^\circ$ PS in the two states.

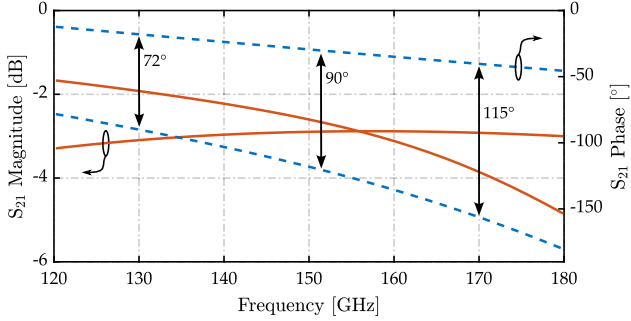


Fig. 6. S_{21} magnitude (solid) and phase (dashed) of the switched Tlines $0^\circ/90^\circ$ PS in the two states.

loss, the other end of $TL_{1,2}$ is terminated by $L_{\text{term}} = 25$ pH which is connected to ground through M_2 . In fact, in the ON-state, $TL_{1,2}$ shunt the input/output ports and are equivalent to a $\lambda/8$ Tline with half the characteristic impedance, $Z_0/2$. L_{term} is sized to show an impedance approximately equal to $+jZ_0/2$, such that a high impedance (ideally an open circuit) appears at the top end of $TL_{1,2}$, thus not affecting the signal transmission. The size of the nMOS switches is set to $W/L = 35 \mu\text{m}/55 \text{ nm}$ as a compromise between IL and phase error, caused by the channel resistance and device parasitic capacitors, respectively. Inductors $L_{p1,2}$ and L_{term} are realized with Tlines. Fig. 6 reports the simulated performance of the $0^\circ/90^\circ$ network with the switches in the ON-state and OFF-state. The relative phase shift is 72° at 130 GHz and 115° at 170 GHz with the IL within 2–3.8 dB. The input/output return loss (not shown) is above 19 dB over the same bandwidth. The core size of the circuit is $230 \mu\text{m} \times 140 \mu\text{m}$.

III. CASCODE STRUCTURE WITH GAIN BOOSTED BY REACTIVE FEEDBACK IN THE CE

The performance of D-band amplifiers in silicon is limited by the low gain of transistors working close to f_{max} , thus suggesting a careful evaluation and selection of the gain stage configuration. The cascode-based structure implemented in the proposed D-band RX front-end is introduced and designed in this section.

The cascode, made by stacking the common-base (CB) and CE HBTs, is largely adopted in D-band amplifiers, enabling higher gain than a CE or CB device alone [25], [26]. Solutions for gain enhancement have also been investigated: in [27]

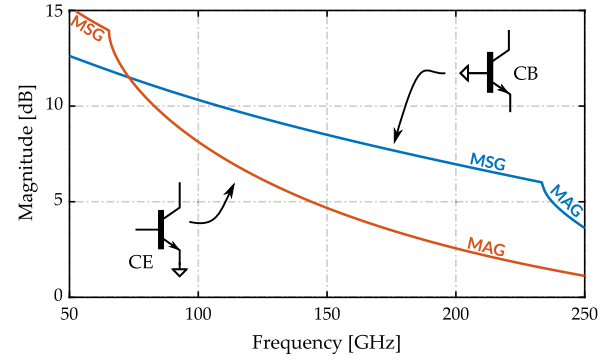


Fig. 7. Maximum power gain of the CE and the CB.

and [28], interstage matching between CE and CB is introduced. In [29], inductive base degeneration at the CB further rises the gain, with the drawback of increasing the already high output impedance and complicating matching networks implementation.

Fig. 7 shows the MAG/MSG curves for the same HBT in CE and CB configurations (emitter area $2 \mu\text{m}^2$ and bias current of 10 mA, giving $f_{\text{max}} = 290$ GHz). It appears evident that the poor performance of the CE device is actually limiting the gain in a cascode structure. Reactive feedback has been investigated as a gain-boosting technique, but it was typically demonstrated on narrowband circuits [30], [31], [32]. Broadband multistage amplifiers with reactive feedback on each stage were also reported, but they rely on stagger tuning of the interstage networks, penalizing gain in favor of bandwidth [33], [34]. In [21], we proposed a two-stage cascode LNA with gain boosted by reactive feedback in the CE. The design flow, specifically focused on low noise amplifiers, appears straightforward and does not penalize the bandwidth. This section extends the technique by elaborating a generalized design flow, finally applied to the gain stage adopted in the proposed D-band RX front-end.

We focus on the highest frequency within the D-band, 170 GHz, where it is more difficult to obtain high gain. The design of a cascode structure starts from the CB device. Looking at Fig. 7, the CB is conditionally stable up to above 200 GHz. In this regime, the maximum stable gain (MSG) gives the maximum achievable power gain if a resistive loss is purposely introduced to ensure unconditional stability [35]. However, the MSG is not an upper limit, and a higher gain can be targeted, provided the source and/or load terminations are selected taking into account stability issues. Fig. 8 reports the power gain (G_P) circles of the CB at 170 GHz, i.e. the loci of load impedance (Z_L) that realize increasing values of G_P from 6 to 8 dB (the MSG of the CB is 8 dB at 170 GHz). The shadowed gray area corresponds to those loads that make the transistor potentially unstable. In principle, a G_P even higher than what considered for the circles in Fig. 8 may be targeted, but Z_L will be closer to the periphery of the Smith chart, i.e. closer the region of potential instability, and with higher nodal quality factor ($Q_n = |X_L|/R_L$, being $Z_L = R_L + jX_L$). Q_n is an important parameter that sets an upper

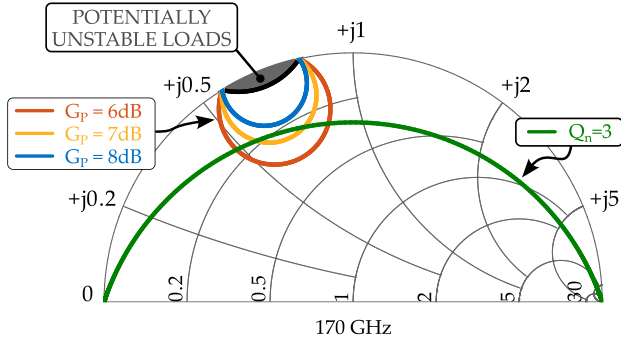


Fig. 8. Power gain and load stability circles of the CB at 170 GHz.

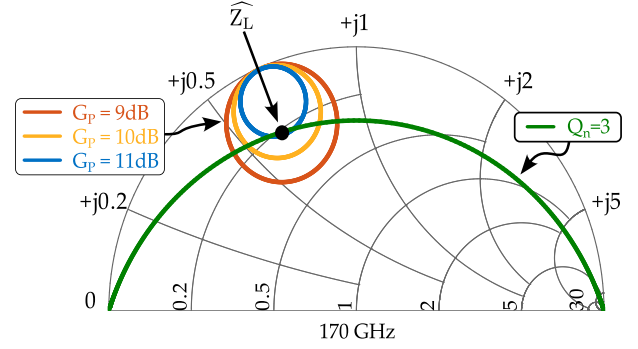


Fig. 10. Power gain circles of the cascode at 170 GHz.

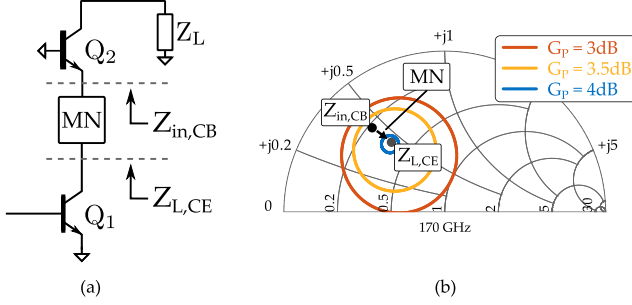


Fig. 9. (a) Schematic of a cascode amplifier. (b) Power gain circles of the CE and impedance levels along the cascode stage at 170 GHz.

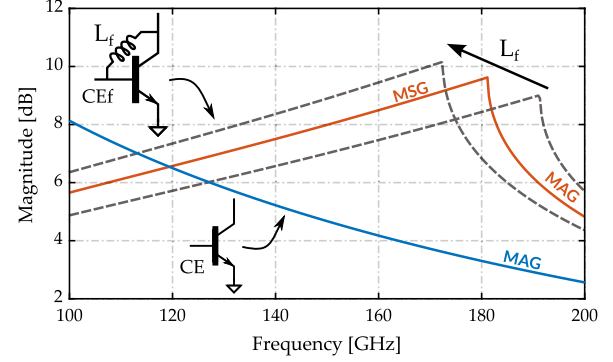
bound on the amplifier bandwidth and also affects the loss of the output matching network, when realized with components featuring a finite quality factor Q (the attenuation of a simple L-type network is $1/(1 + Q_n/Q)$ [36]). Since an excessive Q_n implies narrower band and higher implementation losses, $Q_n = 3$ is set as upper bound. The green curve on the Smith chart in Fig. 8 corresponds to the impedance locus at $Q_n = 3$. The intersection with the G_p circles identifies the required Z_L for the CB and points out a maximum G_p of nearly 7 dB. Once Z_L of the CB is set, we can move to evaluate the performance of the CE device. First, we consider the case of a conventional CE. Then reactive feedback for gain enhancement is discussed.

A. Cascode With a Simple CE

Fig. 9(a) shows a cascode stage formed by a CB staked onto a simple CE. From Fig. 7, the CE is unconditionally stable in D-band. As a consequence, it cannot provide a G_p greater than MAG. At 170 GHz, $G_p = 4$ dB is achieved with output conjugate match.

The power gain circles of the CE are plotted on the Smith chart in Fig. 9(b), together with the impedance seen at the emitter of the CB, $Z_{in,CB}$. Interestingly, with the chosen HBT size and bias current, the CE and CB devices are almost intrinsically matched, being $Z_{in,CB}$ very close to the center of the G_p circles of the CE. For this reason, the interstage network [MN in Fig. 9(a)] performs only a mild impedance transformation of $Z_{in,CB}$ into $Z_{L,CE}$.

The simple cascode stage is now designed and its G_p circles are reported in Fig. 10, together with the $Q_n = 3$ impedance

Fig. 11. Maximum power gain of the standard CE and the CE with reactive feedback (CE_f) at different L_f values (5 pH step). The red curve is for the selected $L_f = 45$ pH.

locus. As expected, the overall gain is the sum of the CE gain (4 dB) and the CB gain. With $Z_L = \widehat{Z}_L$ selected to satisfy $Q_n = 3$, the maximum G_p is 11 dB.

B. Cascode With Reactive Feedback on the CE

From the previous analysis, the CE is limiting the cascode gain since, in the D-band, it is working in the unconditionally stable region and MAG is an upper bound for gain. A way to break this limitation is to apply feedback to shift the CE unconditional stability corner to a higher frequency. Fig. 11 reports the MAG/MSG curves of a CE with reactive feedback (referred to as CE_f), implemented with a small inductor (L_f). The stability corner is shifted across frequency by selecting the value of L_f . With $L_f = 45$ pH, the stability corner falls at 180 GHz, and the device is conditionally stable with unbounded power gain across the full D-band.

Fig. 12 shows the CE_f device in a cascode stage and shows plots of the G_p circles and the load stability circles on the Smith chart for the CE_f with $L_f = 45$ pH. The interstage network between the CE_f and the CB [MN in Fig. 12(a)] transforms the input impedance of the CB ($Z_{in,CB}$, unvaried with respect to the previous case) into $Z_{L,CEf}$, the load for the CE_f . Selection of $Z_{L,CEf}$ sets the CE_f power gain but, looking at the G_p circles in Fig. 12(b), a tradeoff between gain and stability margin exists, like for the CB device previously analyzed, because higher gain needs $Z_{L,CEf}$ to be closer to the region of potential instability. Moreover, when pushing the CE_f toward higher gain, its input impedance (at the base terminal)

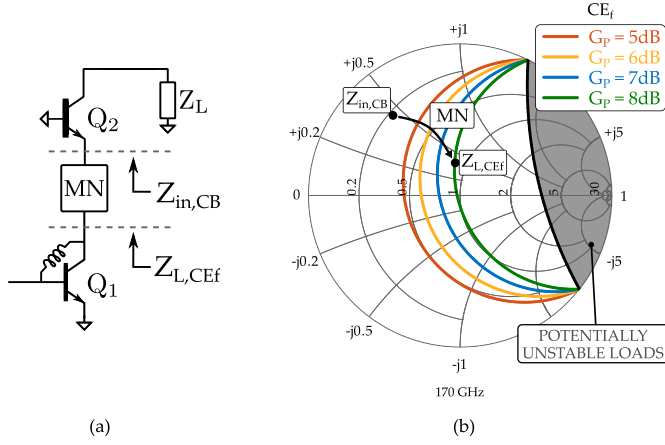


Fig. 12. (a) Schematic of the gain-boosted cascode stage. (b) Power gain and load stability circles of the CE with reactive feedback at 170 GHz and impedance levels along the gain-boosted cascode stage.

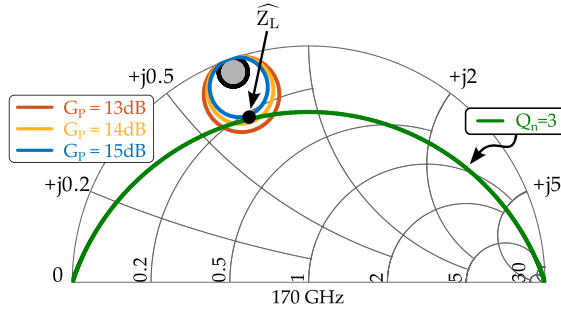


Fig. 13. Power gain circles of the gain-boosted cascode stage at 170 GHz.

and the nodal quality factor, Q_n , increase as well, complicating input matching and limiting the amplifier bandwidth. As an example, moving from $G_p = 5\text{ dB}$ to $G_p = 8\text{ dB}$, Q_n rises from 0.5 to 1. As a compromise between the above issues, G_p of 7–8 dB is selected with the $Z_{L,CEf}$ marked by the black dot in Fig. 12(b).

Finally, Fig. 13 shows plots of the G_p circles at 170 GHz of the cascode with reactive feedback on the CE. The green curve on the Smith chart is again the impedance locus at $Q_n = 3$. Comparing with the conventional cascode stage, whose G_p circles are plotted in Fig. 10, with the same load impedance, $\widehat{Z}_L$, the power gain is increased from 11 to 15 dB.

IV. FRONT-END DESIGN

The D-band RX front-end comprises a two-stage LNA followed by a PS, realized by cascading the passive networks presented in Section II, interleaved by gain stages. The design of the LNA and PS is described in this section.

A. Low-Noise Amplifier

Fig. 14 shows the schematic of the implemented two-stage LNA. All the transistors, with emitter area of $2\text{ }\mu\text{m}^2$, are biased with a current of 10 mA. Fig. 15 reports $\text{NF}_{\min}$ and $f_{\max}$ for a transistor in the CE_f configuration, with $L_f = 45\text{ pH}$, showing that the selected bias point gives simultaneously minimum $\text{NF}_{\min}$ and $f_{\max}$ close to the peak. The cascode

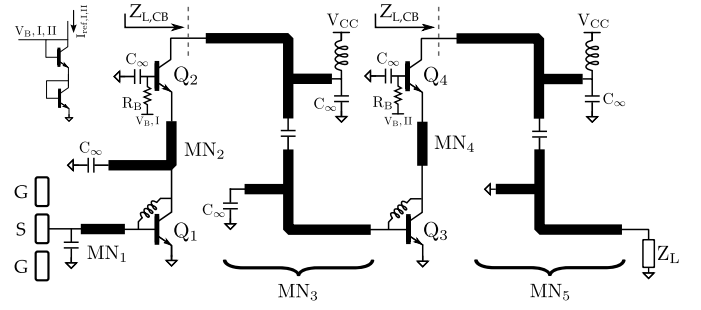


Fig. 14. Schematic of the gain-boosted two-stage LNA.

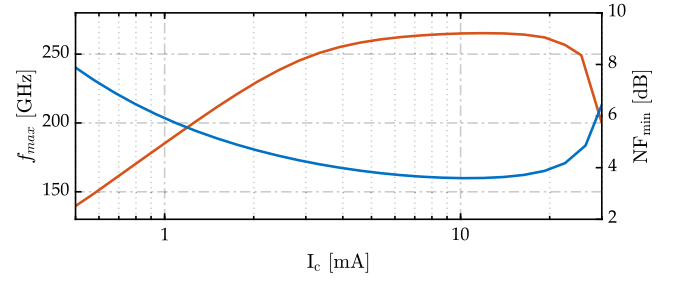


Fig. 15. $f_{\max}$ and $\text{NF}_{\min}$ of the CE_f with $L_f = 45\text{ pH}$ versus collector current ($A_e = 2\text{ }\mu\text{m}^2$).

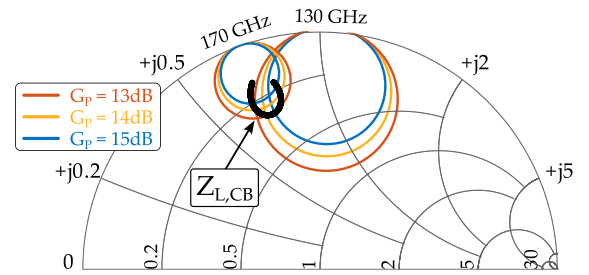


Fig. 16. Power gain circles of the gain-boosted cascode and load impedance synthesized by a fourth-order broadband matching network over the D-band.

structure presented in Section III is used in the second stage ($Q_3 - Q_4$) with $Z_{L,CB}$ the impedance synthesized by the matching network MN_5 as load impedance for Q_4 . With the target of wideband operation, the design starts by considering the G_p circles of $Q_3 - Q_4$ at 130 and 170 GHz, shown in Fig. 16. To have a fairly flat gain over frequency, MN_5 is required to scale up the impedance presented by the cascaded block (Z_L in Fig. 14, nominally $50\text{ }\Omega$) to the impedance $Z_{L,CB}$ at the intersection of the G_p circles at the two frequencies. MN_5 is a fourth-order network with two parallel resonance frequencies, enabling broadband impedance transformation. At RF and lower frequency, such kind of response is commonly achieved with a doubly tuned transformer, realized by magnetically coupling spiral inductors [37]. On the other hand, their accurate modeling is difficult in the D-band and parasitic elements limit self-resonance frequency and quality factor. Tlines are more attractive due to lower loss and a simpler scalable model which limits the iterations with a time-consuming electromagnetic simulator to reach the desired performance. The network MN_5 is realized with a combination of two T-shaped Tline-based sections, separated by a series

capacitor that provides the required AC coupling. The network mimics the response of a doubly tuned transformer and is sized following the design flow described in [38]. The Tlines are realized in the topmost 9th metal layer and show a quality factor $Q_{TL} = 35$ at 165 GHz. The simulated $Z_{L,CB}$ at the input of MN₅, from 100 to 200 GHz, is plotted with the black solid line on the Smith chart in Fig. 16.

Looking again at the schematic of the LNA, the interstage network MN₃ has the same structure of MN₅ and transforms the relatively low impedance at the base of Q₃ to $Z_{L,CB}$ seen at the collector of Q₂. The first stage of the LNA (Q₁–Q₂) is still a cascode structure, but the matching networks MN₁ and MN₂ are sized for optimal noise performance, rather than for maximum power gain. MN₁ transforms the OFF-chip source termination (nominally 50 Ω) into the optimal impedance that minimizes the noise figure of Q₁. MN₂ transforms the impedance at the collector of Q₁ into the optimal impedance for noise of Q₂. This design choice minimizes simultaneously the noise figure of the first two transistors, which account for most of the signal-to-noise ratio degradation within the LNA, with an acceptable penalty in the power gain.

Capacitors labeled C_∞ in the schematic of Fig. 14 provide AC ground paths and are realized with a stack of MIM and MOM components. The supply voltage V_{CC} is fed through low-Q multturn coils to cut uncontrolled signal return paths. Q₁ and Q₃ are diode-connected in DC, and thus, the active stages are biased with a replica-bias circuit, shown in Fig. 14. The voltages generated by the current mirrors are fed to the bases of Q₂ and Q₄ through resistors $R_B = 300 \Omega$. The low-frequency pole that arises with C_∞ fully suppresses the noise of the bias network. From post-layout simulations, with 20 mA DC current from a 2 V supply, the LNA provides 22 dB gain with 5–6.5 dB NF and 0 dBm OP_{1dB}.

B. 0°–360° Phase Shifter

The 0°–360° programmable PS is realized by properly combining the passive networks described in Section II. While the tunable bandpass filters allow a fine phase control resolution, the IL would be prohibitively large if many cells are cascaded to cover the required 360° relative phase change. The PS is thus implemented by cascading the 0°/180° and 0°/90° networks with the minimum number of bandpass filters required to cover the 360° range, with some margin, across the band of interest. Considering the simulation results in Section II, with five tunable bandpass filters (in addition to the 0°/180° and 0°/90° networks), the overall chain covers 0°–375° at 130 GHz and 0°–515° at 170 GHz.

An effortless implementation of the PS in the front-end is by straightly cascading all the passive networks after the LNA, as depicted in Fig. 17(a), and compensate the loss with a post amplifier. The maximum IL of each network, estimated from the simulations in Section II, is reported in the block diagram. Summing all the contributions, the total IL, of 24 dB in the worst case, would have a detrimental impact on the front-end noise. In fact, considering the simulated LNA performance, $NF \approx 6$ and 22 dB gain, the NF after the passive PS would rise to 7.5 dB with an overall signal attenuation of 2 dB.

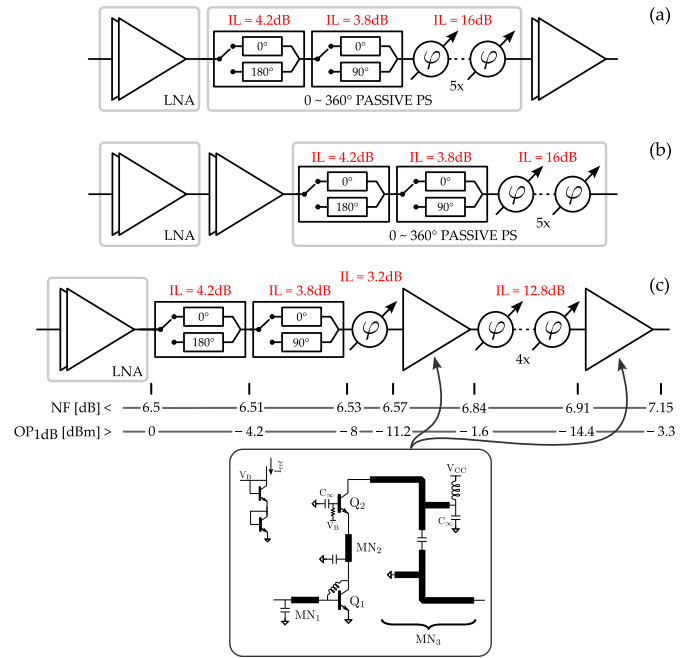


Fig. 17. (a), (b) Different combinations in the cascade of an LNA, a passive PS, and amplifiers. (c) Proposed solution with lossy blocks interleaved with amplifiers. The corresponding worst case NF and OP_{1dB} at the output of each block is reported. Amplifiers are based on cascode stages with gain boosted by reactive feedback on the CE.

Assuming, optimistically, 6 dB NF also for the post-amplifier, the overall NF would rise above 10.5 dB. The noise penalty can be avoided shifting the additional amplifier before the passive PS chain, as shown in Fig. 17(b). However, in this case the linearity is dramatically impaired. In fact, assuming 0 dBm output-referred 1 dB compression point (OP_{1dB}) for the cascaded amplifiers, the 24 dB IL added at the output would limit the OP_{1dB} of the front-end to -24 dBm only.

The above discussion highlights the noise and linearity tradeoff, well-known since the early radio implementations. Remarkably, the same tradeoff afflicts also systems that use an active PS. However, using different PS blocks, it is possible (or at least easier) to split the lossy passive networks and interleave them with active stages, giving higher flexibility to address and mitigate the tradeoff. To this purpose, the architecture in Fig. 17(c) is finally implemented. All the required passive networks are split into two sections, of nearly equal IL, and each one is followed by a single-stage amplifier, balancing the NF and linearity degradation. Each gain stage, with schematic shown in the bottom part of Fig. 17, is the same cascode presented in Section III-B and implemented in the second stage of the LNA. The dual-resonance matching network MN₃ scales up the characteristic impedance of the passive networks to the load required at the collector of Q₂ to have a flat and wideband gain profile. The impedance at the base of Q₁ is already close to 50 Ω , and a simpler network (MN₁) is used at the input. With a bias current of 10 mA from 2 V supply, each gain stage displays $G_P = 13$ dB, $NF = 6.5$ dB, and OP_{1dB} = 1 dBm. The noise and compression performance estimated along the chain, in the worst case, are reported in the block diagram of Fig. 17(c). The dynamic range degradation of the PS is marginal, being the NF and

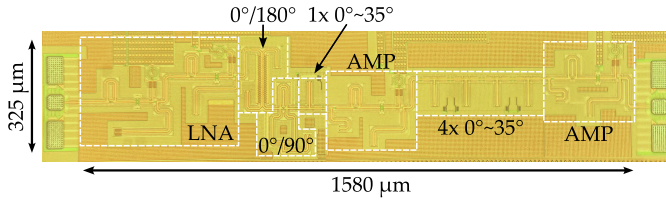


Fig. 18. Chip photograph of the realized D-band RX front-end.

$OP_{1\text{ dB}}$ of the RX front-end only slightly worse than those at the output of the LNA. The test chip also includes a serial register to load the digital configurations required to program the desired phase shifts. The IL variation in the passive networks is compensated by adjusting the bias current of the gain stages, around the nominal value, with an off-chip D-to-A converter.

V. EXPERIMENTAL RESULTS

The RX front-end is realized in STMicroelectronics' SiGe BiCMOS 55 nm technology. Fig. 18 presents the chip photograph. With a rectangular footprint of $1580 \times 325 \mu\text{m}^2$, the chip size, as further commented later, is comparable or lower than previously reported ICs for the same application. The height, less than half the wavelength, easily allows the integration of the front-end into a 1-D phased array. Regarding the implementation of 2-D arrays, it has been recognized that IC dimensions are not scaling according to the increase in operating frequency, still requiring the development of appropriate distribution networks and packaging solutions [39], [40]. For measurement purposes, the chip is glued onto a PCB which provides the supply, biasing, and digital signals through bond-wires. The test setup also includes a custom PCB equipped with a microcontroller, used to program the on-chip serial register, and 5-bit current DACs that set the bias currents of the integrated amplifiers. The small-signal measurements are captured with an Agilent E8361C VNA equipped with VDI WR6.5-VNAX frequency extension modules which are connected to Infinity Waveguide GSG probes. The S-parameter curves are acquired by an automated routine for all the phase shift program codes. To compensate for gain variation at different settings of the passive cells, the bias currents of the amplifiers are swept up to 40% below the nominal value, as a mean to perform variable gain control, except for the last stage, which is kept at the optimal current not to degrade excessively the compression performance. This solution avoids the use of a dedicated variable-gain amplifier. In fact, the overall required gain control range is roughly 9 dB, and the gain variation in 3 dB per stage is easily achieved without altering significantly the impedance match between stages. Fig. 19 shows plots of all the acquired S_{21} (magnitude and phase) curves in gray. Measurements are then processed, with an offline calibration procedure, a subset of coefficients is selected to have a constant-gain 0° – 360° phase shift at three frequencies, 137, 151, and 165 GHz. The corresponding S_{21} curves are highlighted in red, yellow, and blue in Fig. 19, demonstrating an average gain of 20 dB (dashed black line) and a 3-dB bandwidth from 130 to 170 GHz. The S_{11} and S_{22}

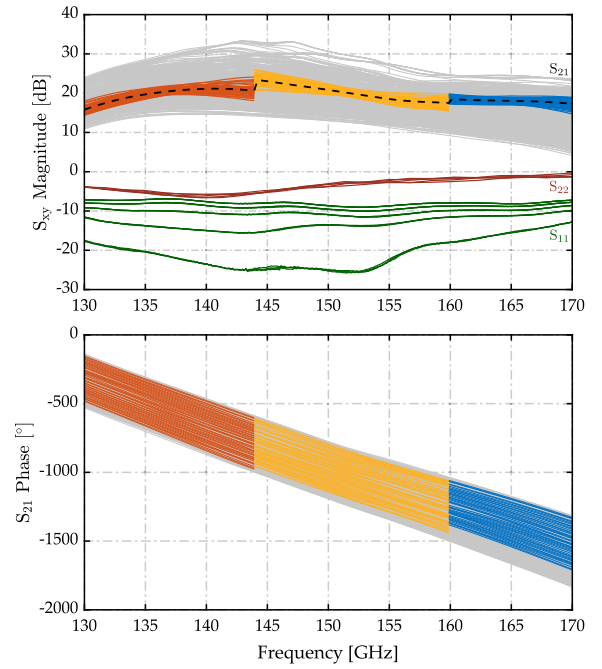


Fig. 19. Measured S-parameters. Highlighted are the curves acquired at the selected calibration settings.

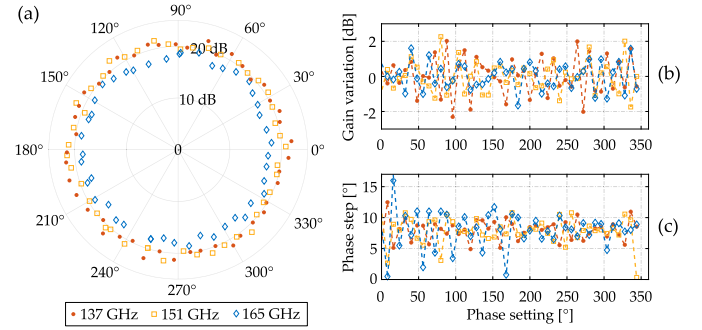


Fig. 20. (a) Measured S_{21} on the polar plot at the frequencies and settings used for offline calibration. (b) Punctual gain variation and (c) phase steps versus phase setting.

curves corresponding to the selected calibration coefficients are visualized in green and purple, respectively.

Fig. 20(a) shows the polar characteristic of the RX front-end using the coefficients calibrated at the three frequencies. The average phase step is 7° . Fig. 20(b) and (c) reports the point-to-point effective phase step, with an error lower than $\pm 7^\circ$, and the gain deviation from the ideal circle which remains within ± 2 dB. The calculated rms gain and phase errors [13] across frequency in the three sub-bands are plotted in Fig. 21 and are on average 1 dB and 10.9° , respectively. The rms phase error is aligned with those reported for PSs based on active vector interpolators [11], [13]. In applications requiring low error, it is convenient to calibrate the front-end at the operating frequency or calibrate the coefficients at more than three frequencies. On the other hand, in large arrays, the phase errors among different antenna elements may be assumed uncorrelated and are thus averaged out [41], [42].

The noise figure is measured with the Y-factor method [43] using a Farran WGNS-06 D-band waveguide noise source,

TABLE I
COMPARISON WITH THE STATE-OF-THE-ART

	This work	[8] RFIC 2020	[5] LMWC 2018	[3] TMTT 2015	[4] SiRF 2017	[9] SSC-L 2021
SiGe Technology	55 nm BiCMOS	130 nm	130 nm	120 nm	130 nm	130 nm
f_t/f_{max}	320/370	300/500	200/220	–/270	250/340	300/500
freq/BW [GHz]	150/40	150/40	115/20*	92/8	125/14	140/60
BW [%]	26.7	27	17	9	11	42.8
Gain [dB]	20	22	13	30	12 [§]	7.3
Phase resolution [°]	7	2	22.5	11	–	20 [‡]
IP _{1dB} [dBm]	–22.5/–21	–22	–17*	–37	–20.7	–30
OP _{1dB} [dBm]	–3.5/–2	–1	–5*	–8	–9.7	–23.7
NF [dBm]	6.8/7.4	10	11*	8 [†]	8 [#]	9 [†]
P _{DC,MAX} [mW]	80	165	53	80	125	436
Area [mm ²]	0.59	0.92	0.82	0.4	0.1	1.02

* simulated value

† T/R switches de-embedded

§ baluns de-embedded

LNA only

‡ estimated from time-delay measures

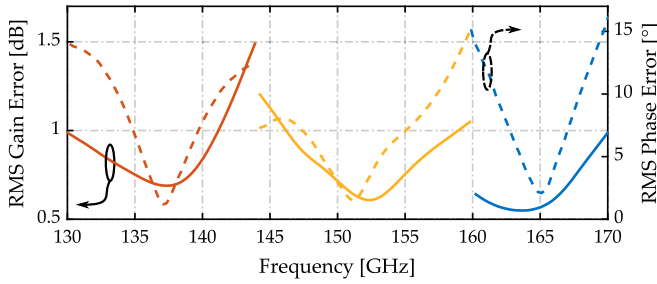


Fig. 21. Measured rms gain (solid) and phase (dashed) errors.

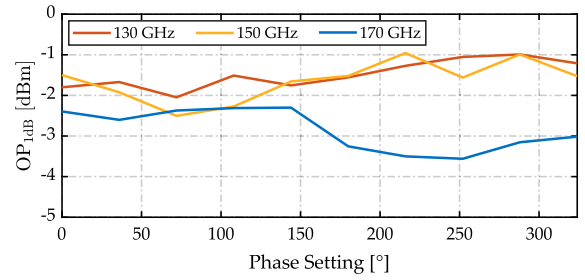
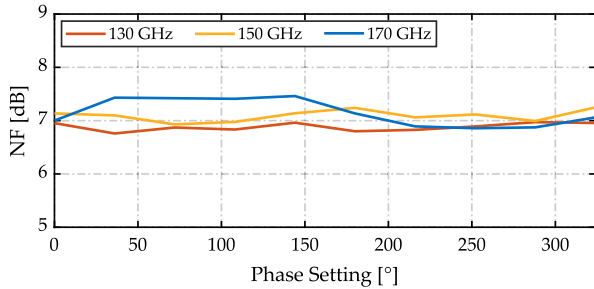
Fig. 23. Measured OP_{1dB} at three frequencies versus phase setting (36° average steps).

Fig. 22. Measured NF at three frequencies versus phase settings (36° average steps).

a VDI WR6.5SAX-M frequency extender (with the high-sensitivity option for enhanced displayed average noise level) and a PXA N9030A spectrum analyzer by Agilent. An additional wideband IF amplifier is inserted after the extender to increase the accuracy of the results. The setup is calibrated at three frequencies (130, 150, and 170 GHz), and measurements are performed programming a 0°–360° phase shift in ten steps (i.e., with average phase steps of 36°) at the three frequencies. The results are plotted in Fig. 22. The NF ranges from 6.8 to 7.5 dB across frequency and at different phase shifts. Gain compression measurements are performed with an ELVA-1 DPM-06 D-band power meter in combination with an Agilent E8257D signal generator and the VDI WR6.5-VNAX. The measured OP_{1dB} at 130, 150, and 170 GHz, still spanning the 0°–360° phase shift in ten steps, is reported in Fig. 23 and, at 130 and 150 GHz, it is from –2 to –1 dBm. OP_{1dB} drops to –3.5 dBm at 170 GHz, for phase shift settings above 150°.

The experimental results are finally summarized in Table I and compared against previously presented RX front-ends designed for similar applications. With center operation frequency, bandwidth, and gain aligned with state-of-the-art, the presented front-end shows the lowest NF. The achieved OP_{1dB} is remarkably higher than other works at comparable power consumption, except [8], which is demonstrating a slightly better compression performance but with twice the power consumption and 2.5 dB worse NF in a technology with higher f_t/f_{max} . A clear advantage of the proposed front-end resides in the higher dynamic range normalized to power dissipation, i.e. the lower NF and higher gain compression point enabled by the use of programmable passive networks to implement the PS. Notably, despite the intensive use of passive networks in this work, the layout footprint is very compact, being the occupied silicon area comparable or lower than the majority of other works.

VI. CONCLUSION

This article presented an RX front-end for D-band phased arrays. The typical configuration comprising a LNA and a PS was analyzed, focusing on the noise and linearity tradeoff. Different passive fine and coarse phase shifting blocks were examined. A discussion on power gain of cascode-based amplifiers followed, recognizing that the CE device is a bottleneck for gain. Reactive feedback was applied to the CE to boost the gain of the cascode amplifier used to build a two-stage LNA. Passive PS blocks and gain-boosted active cores were interleaved to implement a PS with enhanced dynamic range. The combination of the LNA and the PS demonstrated

state-of-the-art noise figure and the highest power efficiency reported to date in similar works, proving the effectiveness of the technique

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